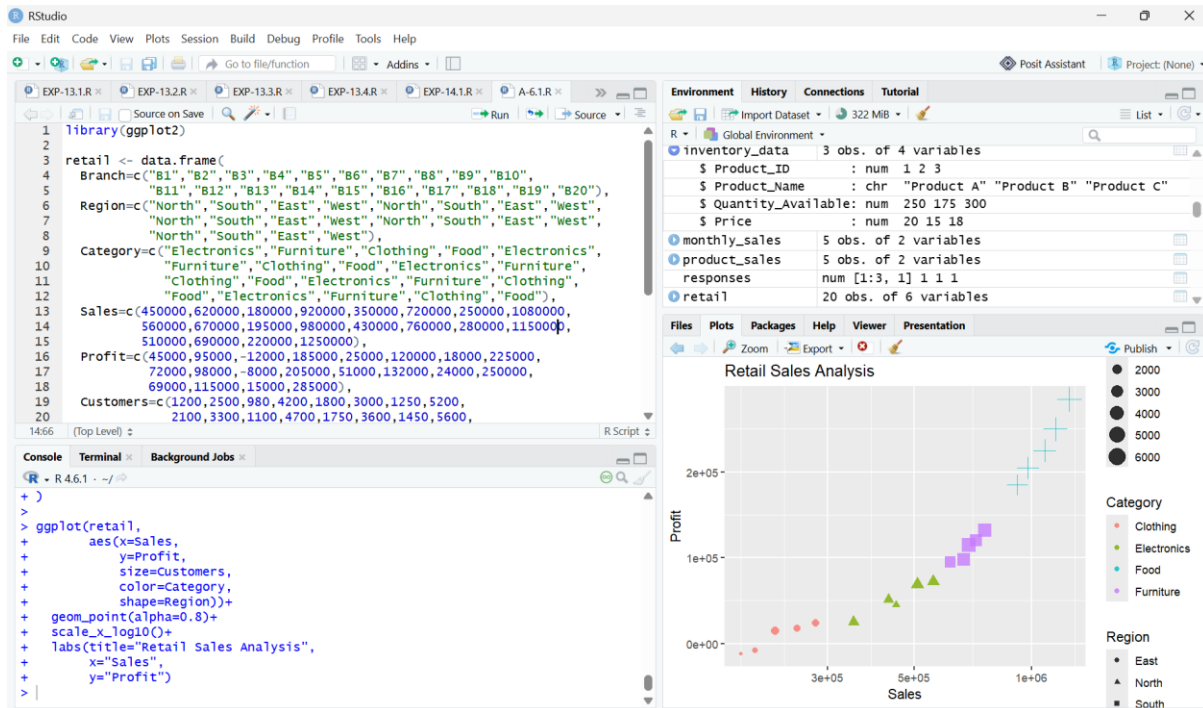


CO1_AT3

QUESTION-1



CODE:

```
library(ggplot2)
```

```
retail <- data.frame(
```

```
  Branch=c("B1","B2","B3","B4","B5","B6","B7","B8","B9","B10",
            "B11","B12","B13","B14","B15","B16","B17","B18","B19","B20"),
  Region=c("North","South","East","West","North","South","East","West",
            "North","South","East","West"),
  Category=c("Electronics","Furniture","Clothing","Food","Electronics",
             "Furniture","Clothing","Food","Electronics","Furniture",
             "Clothing","Food","Electronics","Furniture","Clothing",
             "Food","Electronics","Furniture","Clothing","Food"),
```

```
  Sales=c(450000,620000,180000,920000,350000,720000,250000,1080000,
           560000,670000,195000,980000,430000,760000,280000,1150000,
           510000,690000,220000,1250000),
  Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000,
           72000,98000,-8000,205000,51000,132000,24000,250000,
           69000,115000,15000,285000),
  Customers=c(1200,2500,980,4200,1800,3000,1250,5200,
              2100,3300,1100,4700,1750,3600,1450,5600,
              2100,3300,1100,4700,1750,3600,1450,5600),
```

```
)
```

```

510000,690000,220000,1250000),
Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000,
72000,98000,-8000,205000,51000,132000,24000,250000,
69000,115000,15000,285000),
Customers=c(1200,2500,980,4200,1800,3000,1250,5200,
2100,3300,1100,4700,1750,3600,1450,5600,
2200,3900,1600,6100)
)

```

```

ggplot(retail,
  aes(x=Sales,
    y=Profit,
    size=Customers,
    color=Category,
    shape=Region))+
geom_point(alpha=0.8)+
scale_x_log10()+
labs(title="Retail Sales Analysis",
  x="Sales",
  y="Profit")

```

2. Aesthetic Mapping

Variable	Aesthetic
Sales	X-axis (Log Scale)
Profit	Y-axis
Customers	Bubble Size
Category	Color
Region	Shape

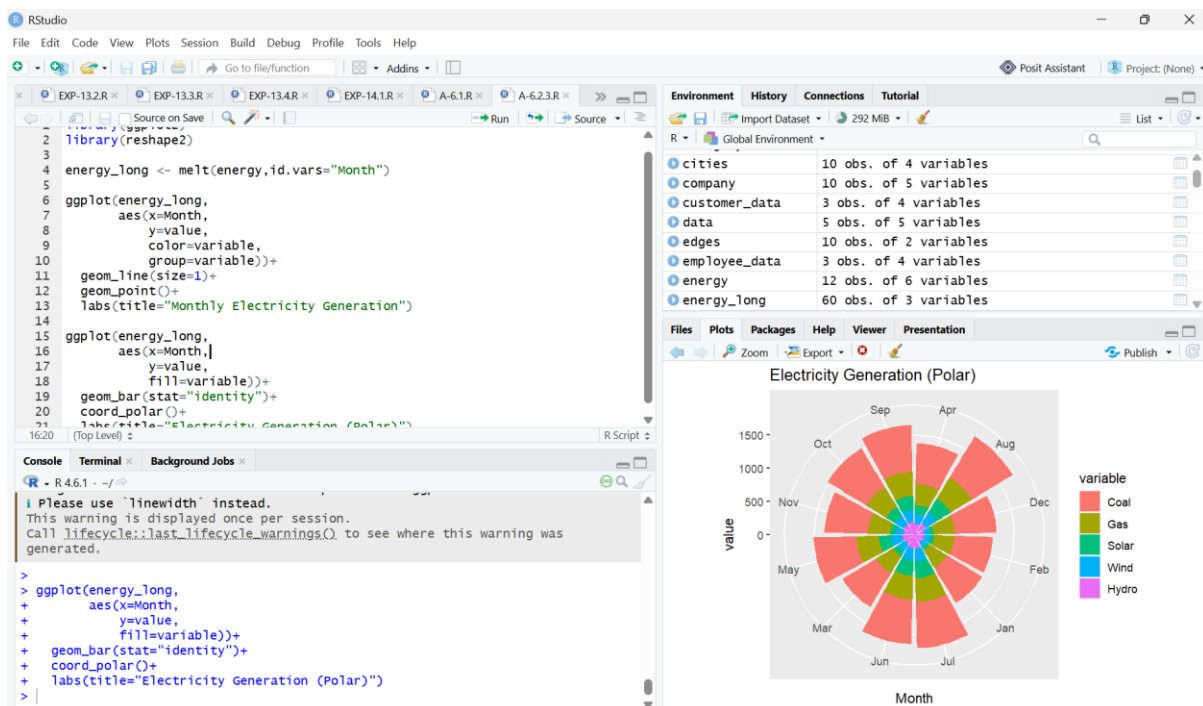
3. Scale Transformation

- **Logarithmic (scale_x_log10())** is used for Sales because the sales values vary widely.
- It spreads data evenly and improves comparison.

4. Justification

- Scatter plot shows relationship between Sales and Profit.
- Bubble size represents customer count.
- Different colors distinguish product categories.
- Shapes separate regions.
- Log scale reduces skewness and improves readability.

QUESTION-2



CODE:

```
library(ggplot2)
```

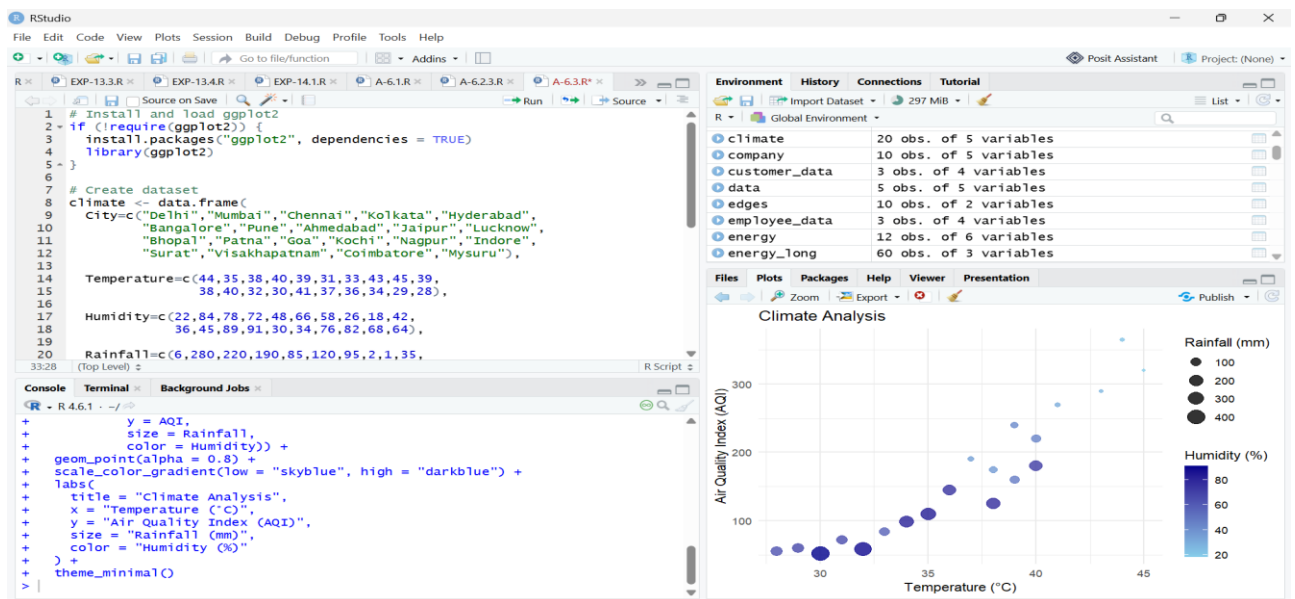
```
library(reshape2)
```

```
energy_long <- melt(energy, id.vars="Month")
```

```
ggplot(energy_long,
  aes(x=Month,
    y=value,
    color=variable,
    group=variable))+
geom_line(size=1)+
geom_point()+
labs(title="Monthly Electricity Generation")
```

```
ggplot(energy_long,
  aes(x=Month,
    y=value,
    fill=variable))+
geom_bar(stat="identity")+
coord_polar()+
labs(title="Electricity Generation (Polar)")
```

QUESTION-3



CODE:

```
# Install and load ggplot2

if (!require(ggplot2)) {
  install.packages("ggplot2", dependencies = TRUE)
  library(ggplot2)
}

# Create dataset

climate <- data.frame(
  City=c("Delhi","Mumbai","Chennai","Kolkata","Hyderabad",
        "Bangalore","Pune","Ahmedabad","Jaipur","Lucknow",
        "Bhopal","Patna","Goa","Kochi","Nagpur","Indore",
        "Surat","Visakhapatnam","Coimbatore","Mysuru"),

  Temperature=c(44,35,38,40,39,31,33,43,45,39,
                38,40,32,30,41,37,36,34,29,28),

  Humidity=c(22,84,78,72,48,66,58,26,18,42,
             36,45,89,91,30,34,76,82,68,64),

  Rainfall=c(6,280,220,190,85,120,95,2,1,35,
             48,72,360,410,12,20,180,240,130,145),

  AQI=c(365,110,125,180,160,72,84,290,320,240,
        175,220,58,52,270,190,145,98,60,55)
)

# Plot

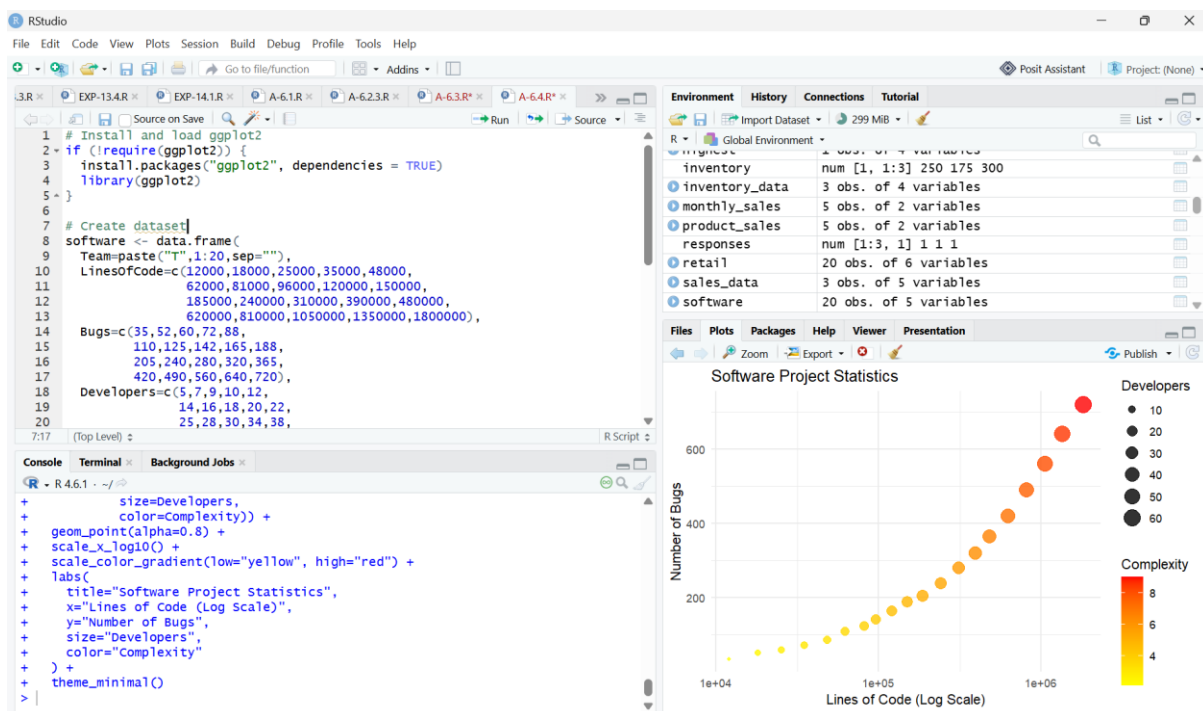
ggplot(climate,
```

```

aes(x = Temperature,
    y = AQI,
    size = Rainfall,
    color = Humidity)) +
geom_point(alpha = 0.8) +
scale_color_gradient(low = "skyblue", high = "darkblue") +
labs(
  title = "Climate Analysis",
  x = "Temperature (°C)",
  y = "Air Quality Index (AQI)",
  size = "Rainfall (mm)",
  color = "Humidity (%)"
) +
theme_minimal()

```

QUESTION-4



CODE:

```
# Install and load ggplot2
```

```
if (!require(ggplot2)) {  
  install.packages("ggplot2", dependencies = TRUE)  
  library(ggplot2)  
}
```

```
# Create dataset
```

```
software <- data.frame(  
  Team=paste("T",1:20,sep=""),  
  LinesOfCode=c(12000,18000,25000,35000,48000,  
    62000,81000,96000,120000,150000,  
    185000,240000,310000,390000,480000,  
    620000,810000,1050000,1350000,1800000),  
  Bugs=c(35,52,60,72,88,  
    110,125,142,165,188,  
    205,240,280,320,365,  
    420,490,560,640,720),  
  Developers=c(5,7,9,10,12,  
    14,16,18,20,22,  
    25,28,30,34,38,  
    42,46,50,55,60),  
  Complexity=c(2.1,2.4,2.7,3.0,3.2,  
    3.5,3.8,4.1,4.5,4.8,  
    5.1,5.4,5.8,6.2,6.6,  
    7.1,7.6,8.0,8.5,9.0)  
)
```

```
# Visualization
```

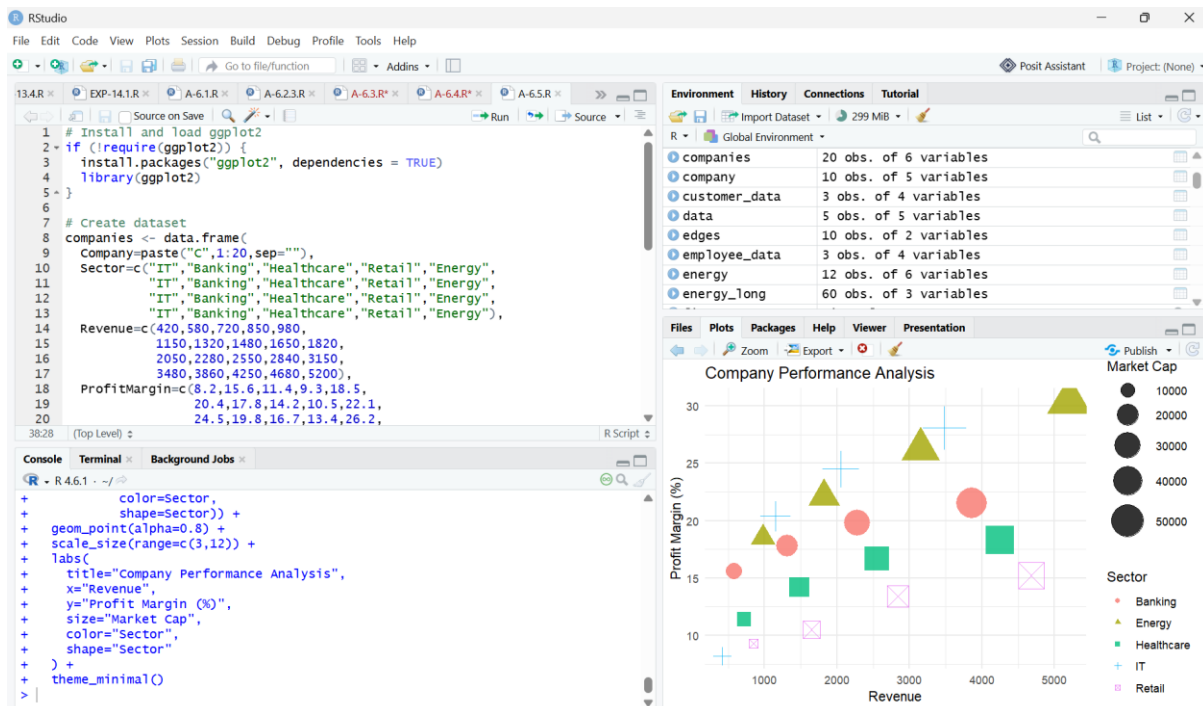
```
ggplot(software,  
  aes(x=LinesOfCode,
```

```

y=Bugs,
size=Developers,
color=Complexity)) +
geom_point(alpha=0.8) +
scale_x_log10() +
scale_color_gradient(low="yellow", high="red") +
labs(
  title="Software Project Statistics",
  x="Lines of Code (Log Scale)",
  y="Number of Bugs",
  size="Developers",
  color="Complexity"
) +
theme_minimal()

```

QUESTION-5



CODE:

```
# Install and load ggplot2
```



```

if (!require(ggplot2)) {
  install.packages("ggplot2", dependencies = TRUE)
  library(ggplot2)
}

# Create dataset
companies <- data.frame(
  Company=paste("C",1:20,sep=""),
  Sector=c("IT","Banking","Healthcare","Retail","Energy",
"IT","Banking","Healthcare","Retail","Energy",
"IT","Banking","Healthcare","Retail","Energy",
"IT","Banking","Healthcare","Retail","Energy"),
  Revenue=c(420,580,720,850,980,
1150,1320,1480,1650,1820,
2050,2280,2550,2840,3150,
3480,3860,4250,4680,5200),
  ProfitMargin=c(8.2,15.6,11.4,9.3,18.5,
20.4,17.8,14.2,10.5,22.1,
24.5,19.8,16.7,13.4,26.2,
28.1,21.5,18.3,15.2,30.4),
  Employees=c(1500,3200,2100,4200,2800,
5100,6800,3900,7200,4500,
8100,9500,6200,10300,7200,
11800,13200,8600,14500,16000),
  MarketCap=c(8500,12000,9800,7600,14500,
18200,21500,16800,13200,24500,
28500,32500,24800,19200,38200,
42500,46800,35600,29800,52000)
)

```

```
# Visualization

ggplot(companies,
       aes(x=Revenue,
           y=ProfitMargin,
           size=MarketCap,
           color=Sector,
           shape=Sector)) +
  geom_point(alpha=0.8) +
  scale_size(range=c(3,12)) +
  labs(
    title="Company Performance Analysis",
    x="Revenue",
    y="Profit Margin (%)",
    size="Market Cap",
    color="Sector",
    shape="Sector"
  ) +
  theme_minimal()
```